

2.4 Counterparty Risk

That leaves one source of risk that we haven't touched upon yet: counterparty credit risk. Each counterparty in a swap enters into the transaction and expects that the other will be able to make the payments that he has committed to make over the life of the transaction. But what if a counterparty, say, the fixed rate payer, should go bankrupt and not have the funds available to make payments in the swap? This would constitute a default on the part of the fixed rate payer. If at the time of default the swap were in favor of the floating rate payer, then the floating rate payer would have a loss if he had to write off the value of the swap due to the insolvency of the fixed rate payer.

Counterparty credit exposure is a term that refers to the amount that someone could lose on a swap (or a portfolio of swaps) if its counterparty defaults. Counterparty credit exposure can be quite significant. Table 2.2 gave us an idea of how much a swap's value can change. Of course, in this table we only considered an instantaneous change in the value of swaps for a given interest rate shift. A credit risk officer will be interested in knowing how much the value of a swap may move over time. One metric that might be proposed is looking at, say, a 95% confidence interval of the value of a swap for each day during the life of the swap. That is, given some model of interest rates that accounts for movements in rates over time, we could come up with a number for each day during the life of the swap for which there is a 95% chance that the value of the swap will not exceed this number. The relationship between the passage of time and the magnitude of the 95% confidence interval is referred to as the *exposure profile* of the swap. Figure 2.5 shows what a typical exposure profile for an interest rate swap looks like.²⁴

Note that the exposure profile is such that when the swap is first executed, it has zero exposure. There are two effects that cause the exposure profile to look the way it does as time goes on. First, as time passes, it is more likely that rates move a lot, and this tends to increase the maximum exposure of the swap. Second, as time goes on, the swap is paying off coupons, and even though rates may be moving a lot from where they were initially, we just don't get a lot of bang for our buck with the now-short maturity swap.²⁵

²⁴The modeling, management, and pricing of credit exposures resulting from OTC derivative trades are complex. Some institutions will actively manage their counterparty credit exposure to a particular counterparty or counterparties. The cost of hedging counterparty credit exposure (i.e., the market value of the counterparty credit exposure) is referred to in the industry as *Credit Valuation Adjustment* (CVA). For further information on these topics, see, for example, Flavell (2010), Cesari et al. (2009), and Ammann (2001).

²⁵Coupons that have come and gone in a swap no longer figure into its existing valuation, much as the dividends already paid by a stock do not contribute to the current value of the stock.

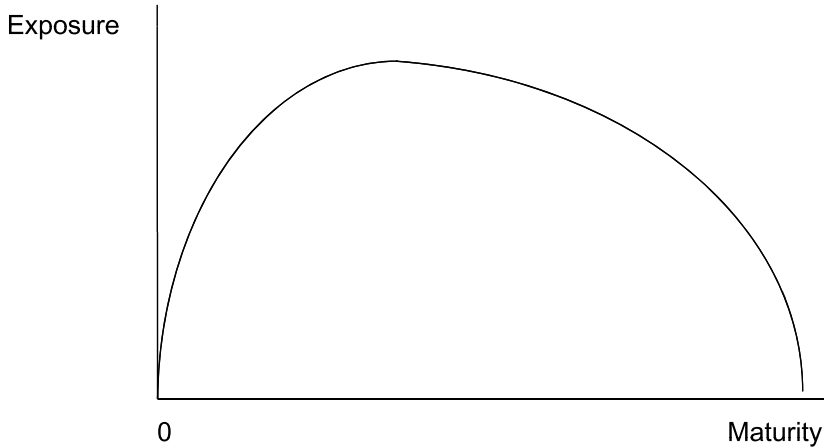


Figure 2.5: Interest Rate Swap Exposure Profile

That is, as time goes on, the remaining swap has a lower PV01. Eventually the second effect dominates the first, and the exposure profile starts coming down. As the swap approaches maturity, the exposure approaches zero.

Cross-currency swaps have a much different exposure profile, as demonstrated in Figure 2.6. Cross-currency swaps have an exchange of principal on the final day of the swap. The spot exchange rate when the swap is first executed is used to set the principal amounts that will be exchanged at maturity. Inasmuch as exchange rates may vary significantly over the life of the swap, the exposure profile on a cross-currency swap remains high even as the swap approaches maturity.

In order to reduce the counterparty risk inherent in swaps, counterparties will often put in place a *Credit Support Annex* (CSA) in addition to an ISDA Master Agreement. The CSA will provide for the transfer of collateral, either cash or securities, from one counterparty to the other in order to immunize against changes in a portfolio of OTC derivatives, including swaps. Under the terms of a CSA, a counterparty, say, a client, who is out-of-the-money on its derivatives portfolio with another counterparty, say, a dealer, will have to post collateral to the latter to protect against the possibility of default. In the event of a default, the dealer could then sell the collateral in the market in an attempt to make himself whole. The terms of a CSA between two counterparties will, among other things, specify the types of collateral that are acceptable, thresholds in the mark-to-market of the derivatives portfolio below which no collateral is posted, and minimum transfer amounts of col-

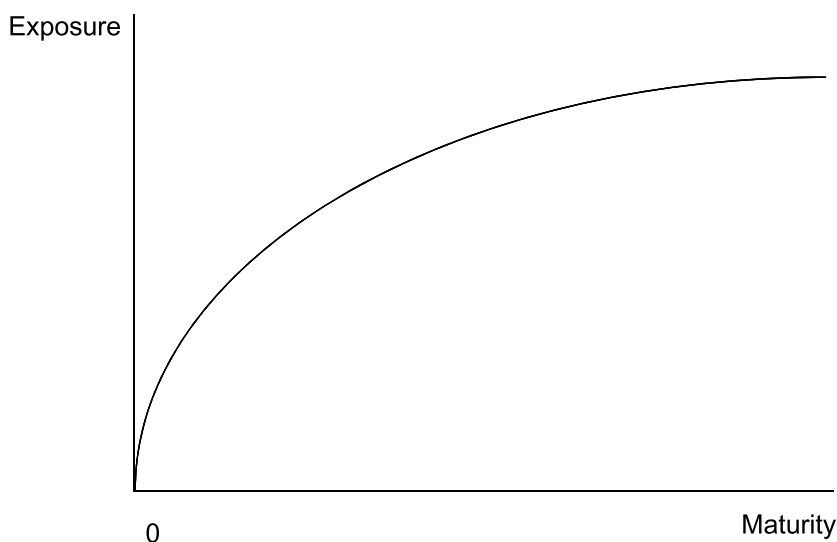


Figure 2.6: Currency Swap Exposure Profile

lateral (preventing nuisance transfer amounts for de minimis changes in the mark-to-market of the derivatives portfolio).²⁶

The Dodd-Frank Wall Street Reform and Consumer Protection Act

The Dodd-Frank Act, signed into law in July 2010 in response to the financial crisis, provides for sweeping changes to the financial system. Although the Act covers a wide range of reform, our focus here is on Title VII, which pertains to the OTC derivatives markets. The goals of the Act as it pertains to derivatives are to increase transparency and reduce systemic risk. The collapse of Lehman and the near-collapse of AIG prompted lawmak-

²⁶One of the most famous examples of the workings of CSAs is found in the bailout of the American International Group (AIG). AIG, with its high credit rating, had enjoyed very advantageous terms (regarding the collateral it was required to post) in the CSAs it had negotiated with many dealers. But many of these CSAs required AIG to start posting additional collateral based on the downgrading of its credit.

AIG was known to have sold credit default swaps on a variety of mortgage-related underlyings. At the height of the crisis, many of these underlyings deteriorated significantly, and the mark-to-market on AIG's credit derivative positions moved sharply against the firm. The downgrades of AIG in conjunction with the continued deterioration of its derivative positions during the crisis triggered collateral calls by many dealers, ultimately leading to a government bailout of AIG in September 2008. See Sorkin (2009) for more details.

ers to implement changes aimed at reducing the risk of systemic failure of the financial system. The concern is that the inability of an insolvent financial institution to perform on its OTC derivatives positions with its various counterparties could cause a ripple effect throughout the financial system.

Title VII of the Act contains three broad mandates. The first of these, known as the *clearing mandate*, requires that certain derivative transactions must be centrally cleared through clearing houses. The U.S. Commodity Futures Trading Commission (CFTC) and the U.S. Securities and Exchange Commission (SEC) are jointly responsible for determining which swaps are to be cleared.²⁷

The second mandate is referred to as the *trading mandate* and requires that any swap that is required to be centrally cleared must be traded on either an exchange or a *swap execution facility*, presuming that the swap is made available for trade. A swap execution facility is defined in Section 721 of the Act as a “trading system or platform in which multiple participants have the ability to execute or trade swaps by accepting bids and offers made by multiple participants in the facility or system.”

The third mandate, the *registration mandate*, requires that certain market participants, known as *swap dealers* and *major swap participants*, register with the CFTC and/or the SEC. A *swap dealer* is defined in the Act to be any person (i.e., an entity) who holds itself out as a dealer in swaps, makes a market in swaps, regularly enters into swaps with counterparties as an ordinary course of business for its own account, or engages in any activity causing the person (i.e., an entity) to be commonly known in the trade as a dealer or market maker in swaps. A *major swap participant* is defined in the Act as any person who is not a swap dealer and maintains a substantial position in swaps, whose outstanding swaps create substantial counterparty exposure that could have serious adverse effects on the financial stability of the U.S. banking system or financial markets or is a financial entity that is highly leveraged relative to the amount of capital it holds.²⁸

It is important to draw a distinction between the clearing of derivative transactions under Dodd-Frank and the trading of derivative transactions

²⁷The CFTC is broadly responsible for the regulation of derivative transactions known as “swaps” in the Act, and the SEC is responsible for “security-based swaps.” A third category of trades known as “mixed swaps” is subject to joint regulation by the CFTC and the SEC. Interest rate swaps in particular fall in the category of “swaps” and as such will be regulated by the CFTC. For our purposes here, we will not distinguish between the various categories.

²⁸For more details on the Dodd-Frank Act, see *The Dodd-Frank Wall Street Reform and Consumer Protection Act*, downloaded from <http://docs.house.gov/rules/finserve/111.hr4173.finsrvcr.pdf>. For a brief summary of the Act, see Davis Polk & Wardwell (2010).

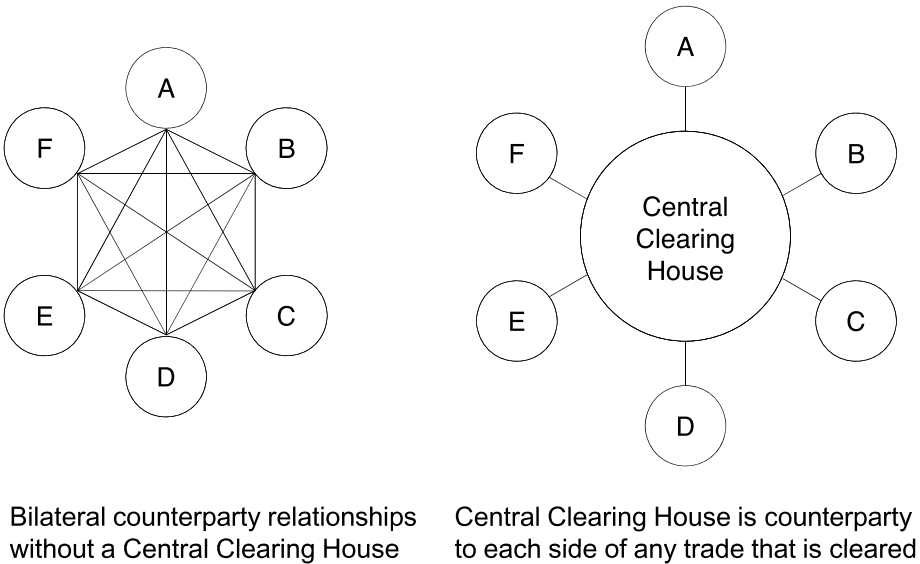


Figure 2.7: The Nature of Counterparty Credit Exposure When Counterparties Face Each Other Bilaterally and Under Central Clearing
 Source: Figure adapted from Duffie, Li, and Lubke (2010).

under Dodd-Frank. The clearing mandate will imply a change from the bilateral nature of counterparty credit risk, which has been synonymous with the OTC derivative markets. This change is depicted in Figure 2.7, which is adapted from a figure presented in Duffie, Li, and Lubke (2010).

The move to central clearing implies that a trade, which may be executed bilaterally or on an exchange, will not have the two counterparties to the trade facing one another directly, but, rather, each counterparty will face a central clearing house.²⁹ The belief held by some is that the existence of central counterparties will mitigate counterparty credit risk and limit the possibility of financial contagion throughout the financial system. In the event of

²⁹As Duffie, Li, and Lubke (2010) point out, Figure 2.7 should not be interpreted as a literal depiction of the move from bilateral counterparties to central clearing. For example, if the counterparties A through F depicted in this figure represent both buy-side institutions and swap dealers, then the left-hand portion of the figure is inaccurate since buy-side institutions typically do not face one another in OTC derivative trades. Moreover, most buy-side institutions will not be *clearing members* of a clearing house themselves and thus will not face the clearing house directly as depicted on the right-hand portion of the figure. Instead, most buy-side institutions will utilize a clearing member or *futures commission merchant* (FCM) to clear their trades.

the insolvency of a large player in the derivative markets, it is proposed by some that central clearing would reduce the risk of triggering a ripple effect. If a clearing member should fail to meet its obligations under its derivative contracts, the belief is that the central clearing house would absorb the losses itself and continue to make payments on derivative transactions to its other members, thereby insulating against a ripple effect.³⁰

Whereas a move of the swaps market to an exchange for purposes of trading would be a radical departure from the market structure currently synonymous with the OTC market, trading swaps on a swap execution facility has the ability to maintain an OTC marketplace. In particular, a swap execution facility may allow counterparties to continue to transact derivatives bilaterally, albeit on a regulated trading system.

2.5 Traditional Uses of Swaps

Having now looked at the risk characteristics of swaps, we will take a look now at a number of traditional ways in which investors use swaps.³¹ In particular, we will examine the use of swaps when corporations issue bonds and want to change the characteristics of the cash flows of the bond. We will also look at asset swaps in which investors buy or sell bonds and use swaps to change the nature of the cash flows of the bond. Finally, we will look at how swaps are used as a part of balance sheet management by a variety of institutions.

2.5.1 New Issue Hedging

Suppose that a corporation wishes to raise \$500mm to undertake an investment project by issuing five-year debt. Given its asset and liability mix, the corporation would like to fund itself on a floating rate basis.³² The corporation works closely with its investment bankers and determines that there is significant investor demand for the corporation to issue one of two alternative bonds. The corporation could issue \$500mm of a five-year floating rate note

³⁰For a more detailed discussion of these and other issues, see Duffie, Li, and Lubke (2010) or Pirrong (2008–2009).

³¹As mentioned in Section 1.1, there are many nontraditional uses of swaps that will be discussed in the chapters ahead.

³²Many corporations choose to fund themselves on a floating rate basis. Doing so prevents a lot of mark-to-market change in the value of debt, which is often desirable. If a corporation has fixed rate liabilities, the value of that liability can move substantially, given a move in rates. When a corporation has a floating rate liability, the value of the liability does not vary tremendously with movements in rates; as rates move, the (expected) coupons on the liability go up or down accordingly, and the value of the liability does not move significantly.

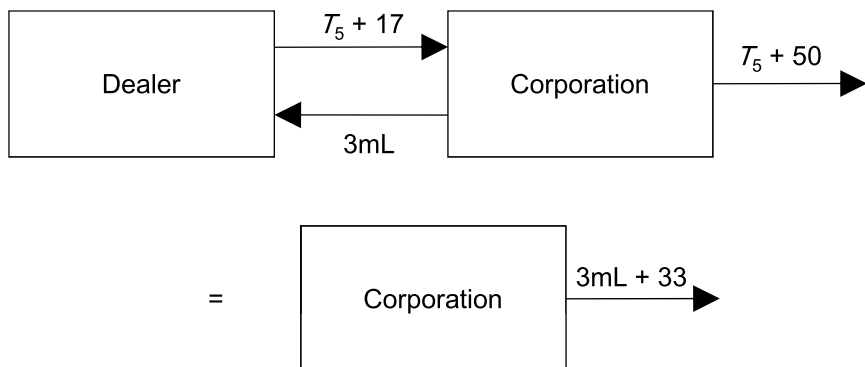


Figure 2.8: New Issue Hedging

with a coupon of 3-month LIBOR + 37 bps. Alternatively, the corporation could issue \$500mm of a five-year fixed rate bond with a coupon of $T_5 + 50$ bps paid semiannually. What should the corporation do if it wants to have floating rate funding and to obtain the most attractive funding possible?

Although the corporation wants to fund itself on a floating rate basis, this does not mean that it should disregard the opportunity to issue a fixed rate bond. Let's go back to Table 1.1. Notice that if a client wants to receive fixed in five years, a dealer is willing to pay $T_5 + 17$. That is, the bid side swap spread in five years is 17 bps, indicating that a dealer will pay a fixed coupon that is 17 bps over the yield of the five-year on-the-run Treasury in exchange for 3-month LIBOR in a five-year swap.

If the corporation were to issue a fixed rate bond with a coupon of $T_5 + 50$ and enter into a fixed-floating swap in which it receives fixed from a swap dealer at $T_5 + 17$, then the corporation would create *synthetic floating rate funding* of 3-month LIBOR + 33 bps. This strategy would save the corporation 4 bps running (i.e., over the life of the transaction) as compared to issuing a floating rate bond with a coupon of 3-month LIBOR + 37 bps. This opportunity is shown in Figure 2.8.

How do we figure out that the corporation would create synthetic floating rate funding of LIBOR + 33 bps if they issue a fixed rate bond and swap to floating? In Figure 2.8, the fixed semiannual 30/360 coupon payments of $T_5 + 50$ that the corporation is paying to the bondholders more or less "cancel out" with the fixed semiannual 30/360 coupon payments of $T_5 + 17$.³³ These flows more or less cancel out because if you do the math of

³³Recall that in a spread trade the Treasuries are traded not to exactly offset the fixed cash flows in the swap but rather to immunize against changes in the value of the swap

these two flows in Figure 2.8, the “outflow” of $T_5 + 50$ partially nets against the “inflow” of $T_5 + 17$, giving $-(T_5 + 50) + T_5 + 17 = -33$ bps. That is, the net of these two flows leaves the corporation paying out an annuity of 33 bps semiannually over the life of the trade. Thus, the corporation is left paying 3-month LIBOR quarterly Act/360 and also paying an annuity of 33 bps, or more simply, 3-month LIBOR + 33 bps.³⁴

So if the corporation issues a fixed rate bond and swaps to floating, it creates synthetic floating rate funding of 3-month LIBOR + 33 bps. Recall that if the corporation had issued a floater, it would have to pay coupons equal to 3-month LIBOR + 37 bps. By issuing fixed and swapping, the corporation is saving itself 4 bps running, or over the life of the transaction. The value of this savings can be computed by approximating the PV Coupon 01 of a 5-year swap. The PV Coupon 01 for \$100mm 5-year swaps is approximated by putting

$$\begin{aligned}\text{PMT} &= 1 \text{ bp}/2 \\ n &= 10 \\ i &= 1.606\%/2\end{aligned}$$

and getting PV = 4.8 bps, or around \$48,000. Thus, the present value of the savings from the synthetic floating rate funding is around³⁵

$$4 \times \$48,000 \times 5 = \$960,000.$$

The factor of 4 in the preceding equation comes from the fact that we are saving 4 bps, and the factor of 5 comes from the fact that the notional on the swap is \$500mm, whereas the PV Coupon 01 was computed on an assumed notional of \$100mm.

The corporation has accomplished a few other things by issuing fixed rate bonds and swapping into floating. First, by transforming the nature of their

that result from movement in rates. In new issuance hedging, this is not the case. Most often the fixed cash flows in the swap are set to exactly offset the cash flows from the bond being issued. Corporations often choose to do this for a number of reasons, not the least of which is to qualify for certain (simplified) accounting treatment of the swap as a hedge.

³⁴It's not so simple, actually. To be technically correct, we can't just move the 33 bps, which are paid semi bond to the floating side that is paid quarterly money. There would be a small adjustment to account for the fact that one leg pays semi bond and the other leg pays quarterly money. In other words, there is an adjustment made both for the change in the frequency and for the change in the day count for the spread. We will ignore this adjustment for our purposes here.

³⁵This calculation is not entirely accurate. The PV Coupon 01 is calculated assuming a 1 bp change in the fixed leg of a swap, which pays semi bond. The 4 bps saving here is on the floating side, which pays quarterly money. Thus, the savings given here is approximately correct.

liability from fixed to floating, the duration of their liability moves to (near) zero. That is, the value of their liability is not going to change significantly when interest rates change. This is not the case with a fixed rate liability; as rates go down, the value of a fixed rate liability increases, and vice versa. Second, if the yield curve is upward sloping, as it often is, then the corporation's interest expense in the current period may be significantly reduced by swapping from fixed to floating. Of course, if the corporation issues a fixed rate bond and swaps to floating, it is exposed if rates subsequently move higher. In this case the interest expense the corporation faces could be higher than if it had maintained its fixed rate liability.

New Issue Hedging Using Cross-Currency Swaps

Let's consider now a second example that shows how a cross-currency swap may be used to hedge an issuance in one currency to create synthetic exposure in another currency. Suppose there is a Japanese corporation that would like to raise 8.53bn yen in order to undertake an investment project. The corporation has decided that it would like to fund itself on a floating rate basis denominated in yen. The term of the investment project is five years.

The corporation works with its investment bankers who inform the corporation that there is demand from investors for the corporation to issue a five-year floater denominated in yen. The bonds would be issued at par and would have a coupon of JPY LIBOR + 25. Alternatively, there is also sufficient demand from investors for the Japanese corporation to issue a five-year floater denominated in USD. The principal amount on this bond would be \$100mm, the equivalent of 8.53bn yen at the assumed spot exchange rate of 85.3. This bond would also be issued at par and would have a coupon of USD LIBOR + 27. Which of the two alternatives should the corporation pursue?

The answer is, it depends. Although the corporation wishes to fund itself on a floating basis denominated in yen, it should not dismiss the possibility of issuing in dollars. The corporation can issue in dollars and use a cross-currency swap in order to swap the coupons to floating yen, creating synthetic floating rate funding denominated in yen. The determination of which alternative is more advantageous depends on the pricing implicit in the cross-currency swap. If the corporation issues a dollar-denominated floater, it will pay coupons of USD LIBOR + 27. Suppose that the corporation contacts a swap dealer and indicates it may have interest in entering into a dollar-yen cross-currency basis swap. The corporation indicates that the terms of the swap would be as follows:

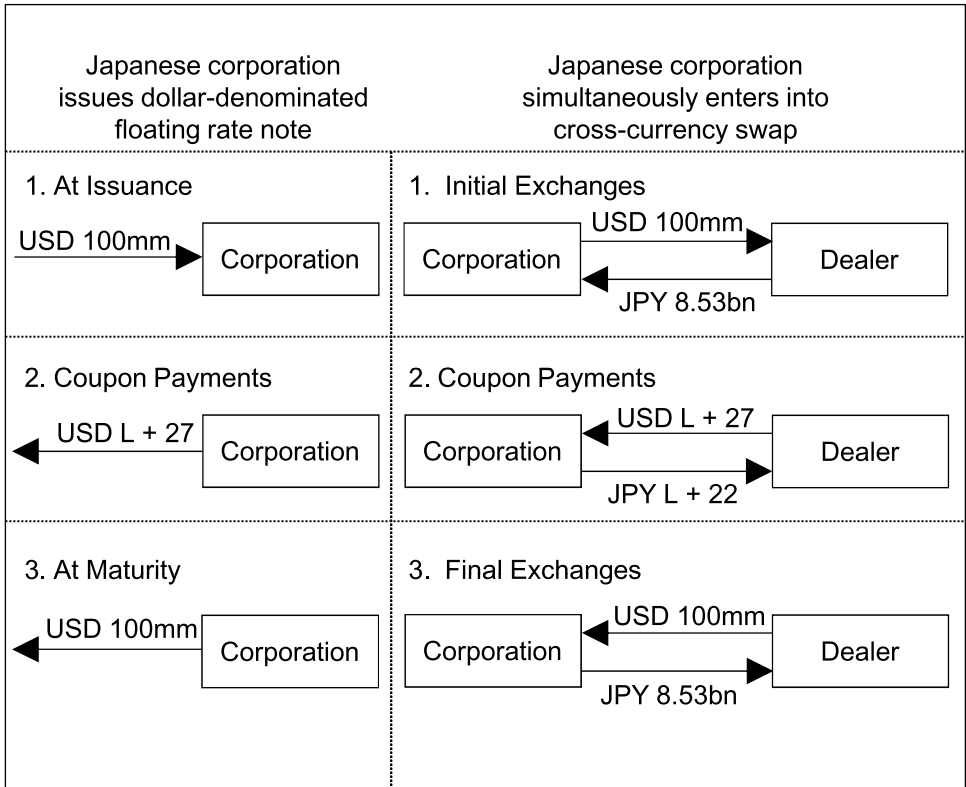


Figure 2.9: Japanese Corporation Issuing in Dollars and Entering into a Cross-Currency Swap in Order to Create Synthetic Floating Rate Funding Denominated in Yen

1. Initial Exchange: The corporation would pay \$100mm to and receive 8.53bn yen from the dealer.
2. Interest Payments: The corporation would receive USD LIBOR + 27 and would pay JPY LIBOR + x to the dealer.
3. Final Exchange: The corporation would receive \$100mm from and pay 8.53bn yen to the dealer.

The spread x will be set such that the cross-currency swap is an NPV = 0 transaction. Suppose that the dealer prices up the swap and determines that $x = 22$ bps. In this case, the corporation is better off issuing in dollars and